JAIN Knowledge Campus, Jayanagar 9th Block, Bengaluru

DEPARTMENT OF COMPUTER SCIENCE AND IT

MCA PROGRAMME

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EXP: 7 - CONNECT POWER BI TO AZURE SQL AND CREATE A DASHBOARD WITH REAL-TIME INSIGHTS

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SUBJECT: CLOUD ANALYTICS-LAB

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1 Aim

To connect Power BI to an Azure SQL Database and create a dashboard that shows student performance insights, with data refreshed from the database rather than from a static file. The experiment uses the student performance data already used in Experiments 04 and 06, so the dashboard can be checked against known values instead of against invented data.

2 Objective and Theory

The objectives are to create an Azure SQL Database, load a real dataset into it through Azure Data Factory (ADF), connect Power BI to that database, build dashboard visuals from the database table, and demonstrate that a database refresh brings updated data into the report.

Azure Data Factory (ADF) is a cloud data integration service for building pipelines that move and transform data [1]. In this experiment, ADF copies a comma-separated values file from Azure Blob Storage into Azure SQL Database. Azure SQL Database is a managed relational database service in Microsoft Azure, and it supports standard Transact-SQL (T-SQL) queries, tables, and aggregations [2]. Power BI is a business analytics service for creating interactive reports and dashboards. DirectQuery is a Power BI connection mode in which visuals query the source database at refresh time, instead of importing a copy of the data into the report file [3]. In the Power BI Service quickcreate flow used for this experiment, no DirectQuery choice was offered, so the report actually used Import mode, as recorded in Step 6. With Import mode, updated data reaches the report through a semantic model refresh, not through a live query at visual load time, and that refresh step is the real-time insight mechanism demonstrated here.

The dataset contains 50 student records with scores in mathematics, reading, and writing, attendance rate, study hours per week, and a final grade from A to F. Two fields have missing values in the source file, attendance rate and study hours per week, each with 45 non-null values out of 50. Those missing values are kept as null values in the database, and averages are calculated over the non-null values, which is stated here so that the averages in Section 5 can be reproduced exactly.

3 Architecture Flow

Data flows from Azure Blob Storage through ADF into Azure SQL Database, and Power BI reads the database table through a semantic model refresh, as shown in Figure 1.1. Table 1.1 lists the resources used, with the same naming pattern as Experiments 01, 04, 05, and 06.

Table 1.1 Resources used in Experiment 07

Component	Resource name	Role
Resource Group	25MCAR0141	Logical container for all experiment resources.
Storage Account	calablake141	Azure Data Lake Storage Gen2, rawdata container, student_performance.csv.
Data Factory	calab-adf-141	Pipeline PL_Exp07_BlobToSql, Copy activity.
SQL Server	calab-sql-141.database.windows.net	Azure SQL Server, Central India, version 12.0.
SQL Database	calab-db-141	Basic, 5 DTU, table StudentPerformance.
Power BI	Power BI Service, Import	Report Exp07_PowerBI_AzureSQL, five visuals, semantic model refresh.

Source: Azure resource list and ADF run record, verified 2026-11-01.

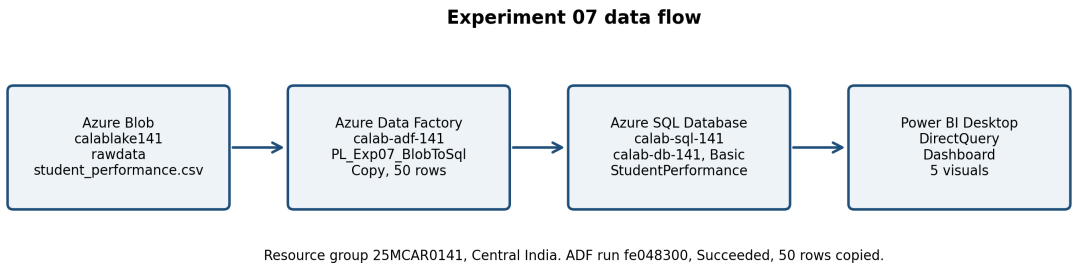


Figure 1.1 Experiment 07 data flow

Blob Storage to ADF to Azure SQL Database to Power BI, with semantic model refresh.

4 Procedure

Step 1 (Azure SQL Database creation). Create SQL Server calab-sql-141 in resource group 25MCAR0141, location Central India, and database calab-db-141 with service objective Basic, 5 DTU, and local backup redundancy. Allow Azure services in the server firewall rule, so that ADF, which runs in Azure, can connect to the database.

Step 2 (ADF linked services and datasets). In Data Factory calab-adf-141, create linked service LS_Exp07_Blob for storage account calablake141 and linked service LS_Exp07_Sql for database calab-db-141. Create dataset DS_Exp07_Blob_CSV for rawdata/student_performance.csv with first row as header, and dataset DS_Exp07_Sql_Table for table StudentPerformance.

Step 3 (ADF pipeline). Create pipeline PL_Exp07_BlobToSql with a Copy activity, CopyBlobToSql, with DS_Exp07_Blob_CSV as source and DS_Exp07_Sql_Table as sink. Add a preCopyScript that creates table StudentPerformance if it does not exist, with the columns listed in create_table.sql in the GitHub folder.

Step 4 (First run and correction). Run the pipeline. The first run failed with Invalid object name StudentPerformance, because the table did not exist and the sink did not create it automatically. Correct the pipeline with the preCopyScript in Step 3 and run it again. The second run succeeded, as recorded in Table 1.2. The failure is reported here because it is part of the execution record.

Step 5 (Verification). Check the ADF activity run output: rowsRead 50, rowsCopied 50, dataRead 3,174 bytes, dataWritten 4,754 bytes, copy duration 12 seconds. These values match the source file, which has 50 data rows plus a header row.

Step 6 (Power BI connection). In Power BI Service, create a report from Azure SQL Database with server calab-sql-141.database.windows.net and database calab-db-141, using Microsoft account authentication, and select table StudentPerformance. The quickcreate flow used offered no DirectQuery choice, so the connection mode actually used was Import, verified in the semantic model settings. Last refresh succeeded on 2026-11-01 at 7:50 PM, and the semantic model refresh section states that a refresh schedule imports data from the data source into the semantic model.

Step 7 (Dashboard visuals). Build five visuals from table StudentPerformance: KPI cards for total students and average scores; a clustered column chart of average mathematics, reading, and writing scores by FinalGrade; a bar chart of student count by FinalGrade; a donut chart of student share by Gender; and a table of FinalGrade, student count, and average scores. The queries for these visuals are in dashboard_queries.sql in the GitHub folder.

Step 8 (Real-time insight demonstration). After building the dashboard, update a value in the database, for example by correcting a score, and refresh the semantic model in Power BI Service. The visual values change after that refresh, because Import mode brings the updated data into the semantic model. Restore the original value after the demonstration, so that the dashboard data in Section 5 remains the submitted record.

5 Proof of Work

The GitHub repository for this experiment is https://github.com/25MCAR0141/Cloud-Analytics-Azure/tree/main/Lab/Exp_07 . The repository contains the README, architecture diagram, dashboard charts, SQL scripts, and ADF pipeline definition, without connection secrets. Table 1.2 records the ADF execution, and Table 1.3 records the dashboard data calculated from the 50 rows in calab-db-141.

Table 1.2 ADF pipeline execution, Experiment 07

Item	Value
Pipeline	PL_Exp07_BlobToSql
Successful run	fe048300-50fb-4bd9-8236-afc838135573, Succeeded, 2026-11-01
Copy activity	CopyBlobToSql, Succeeded
Rows read / copied	50 / 50
Data read / written	3,174 bytes / 4,754 bytes
Copy duration	12 seconds, 4 Data Integration Units
Failed first run	11462f87-e833-4595-8796-f7b30d2449cc, Invalid object name StudentPerformance, corrected with preCopyScript

Source: ADF pipeline run and activity run records, 2026-11-01.

Table 1.3 Dashboard data, average scores by FinalGrade

FinalGrade	Students	Avg MathScore	Avg ReadingScore	Avg WritingScore
A	16	87.50	90.38	88.62
B	12	73.17	76.58	74.83
C	10	57.60	53.80	53.10
D	6	46.67	43.67	42.17
F	6	38.67	34.00	36.33

Source: calab-db-141, table StudentPerformance, 50 rows, verified 2026-11-01. Overall averages are MathScore 67.32, ReadingScore 67.38, and WritingScore 66.36.

Figure 1.2 shows the average scores by FinalGrade from Table 1.3, and Figure 1.3 shows the student count by FinalGrade. These charts are plotted from the same 50 rows in calab-db-141 that Power BI imported into its semantic model, and they match the visuals actually built in Power BI Service in Step 7.

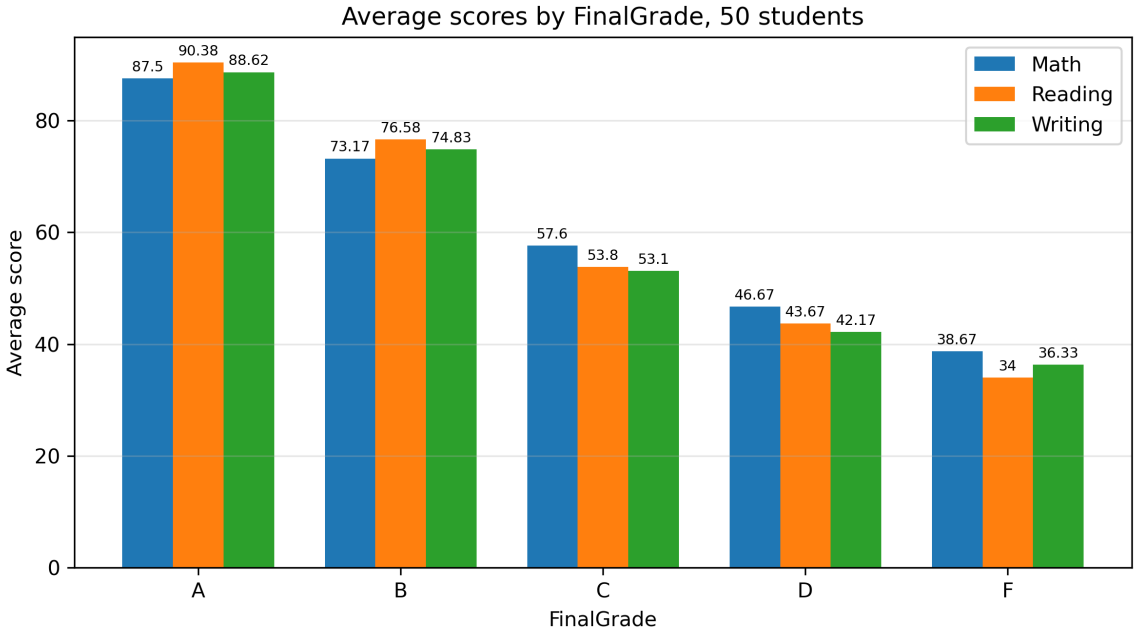


Figure 1.2 Average scores by FinalGrade

Mathematics, reading, and writing averages fall consistently from grade A to grade F.

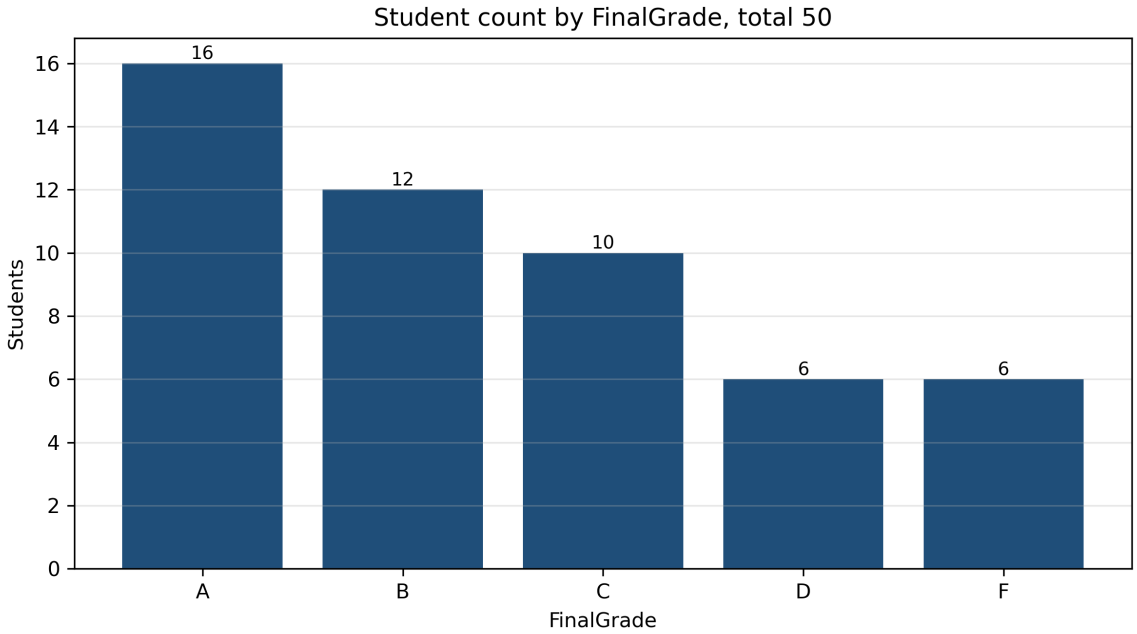


Figure 1.3 Student count by FinalGrade

Grade A has the largest group, 16 students, and grades D and F have the smallest groups, 6 students each.

Power BI report record. The saved Power BI Service report is Exp07_PowerBI_AzureSQL, at https://app.powerbi.com/groups/cfa9da9e-3370-4b81-b46c-b14de945b284/reports/465564ae-997f-4257-8264-fb4c77037e35/d2678006950ecc77ae81?experience=power-bi , with semantic model Exp07_PowerBI_AzureSQL,

refreshed 2026-11-01 at 7:50 PM. Page 1 contains five verified visuals: KPI cards for student count 50 and average scores 67.32, 67.38, and 66.36; a clustered column chart of average scores by FinalGrade; a clustered bar chart of student count by FinalGrade; a donut chart of student share by Gender, Female 25 and Male 25; and a table of FinalGrade, student count, and average scores, with total 50, 67.32, 67.38, and 66.36. No Power BI screenshot has been fabricated for this record.

Cost note. Azure SQL Database calab-db-141 is Basic, 5 DTU. Keep the database only until the Power BI Desktop run and submission are complete, then delete the database and, if no other experiment needs it, the server, to stop the hourly charge.

References

[1] Microsoft, “Azure Data Factory documentation,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/data-factory/

[2] Microsoft, “Azure SQL Database documentation,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/azure-sql/database/

[3] Microsoft, “Use DirectQuery in Power BI Desktop,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/power-bi/connect-data/desktop-directquery-about